

PRODUCT CIRCUIT DIAGRAM

A SYSTEMS VIEW OF HOW VALUE FLOWS THROUGH A PRODUCT — REVEALING FRICTION, MISMATCH, AND OPPORTUNITY

“Utilities are sort of impedance-matched with the government. If you look at a utility, they are heavily regulated by public utility commissions. So they’re sort of impedance-matched with the government—both literally and figuratively” —Elon Musk

🧠 CORE ANALOGY

Electrical Concept	Symbol	Product/system Equivalent
Voltage (V)	⚡⚡⚡	Strength of User Intent / Business Demand
Current (I)	—	Flow of Value / progress
Resistance (R)	—W—	Friction (UX, process, org, data)
Impedance / Mismatch	└┐	System Misalignment (handoffs,formats etc.)
Capacitor / Delay	— — —	Waiting / Buffering (queues, delays)
Amplifier	▶	Simplify / Automate / AI
Blackbox	■	Stable, trusted subsystem
Whitebox	□	Problem area to inspect
Measurement	🕒	Outcome (time, cost, success rate)

⚡ CREATING VALUE ⚡

Outcome Quality ∝ (Flow - Resistance - Mismatch - Delay) + Amplification

THE UNLOCK 🔒

Instead of asking:

“What feature should we build?”

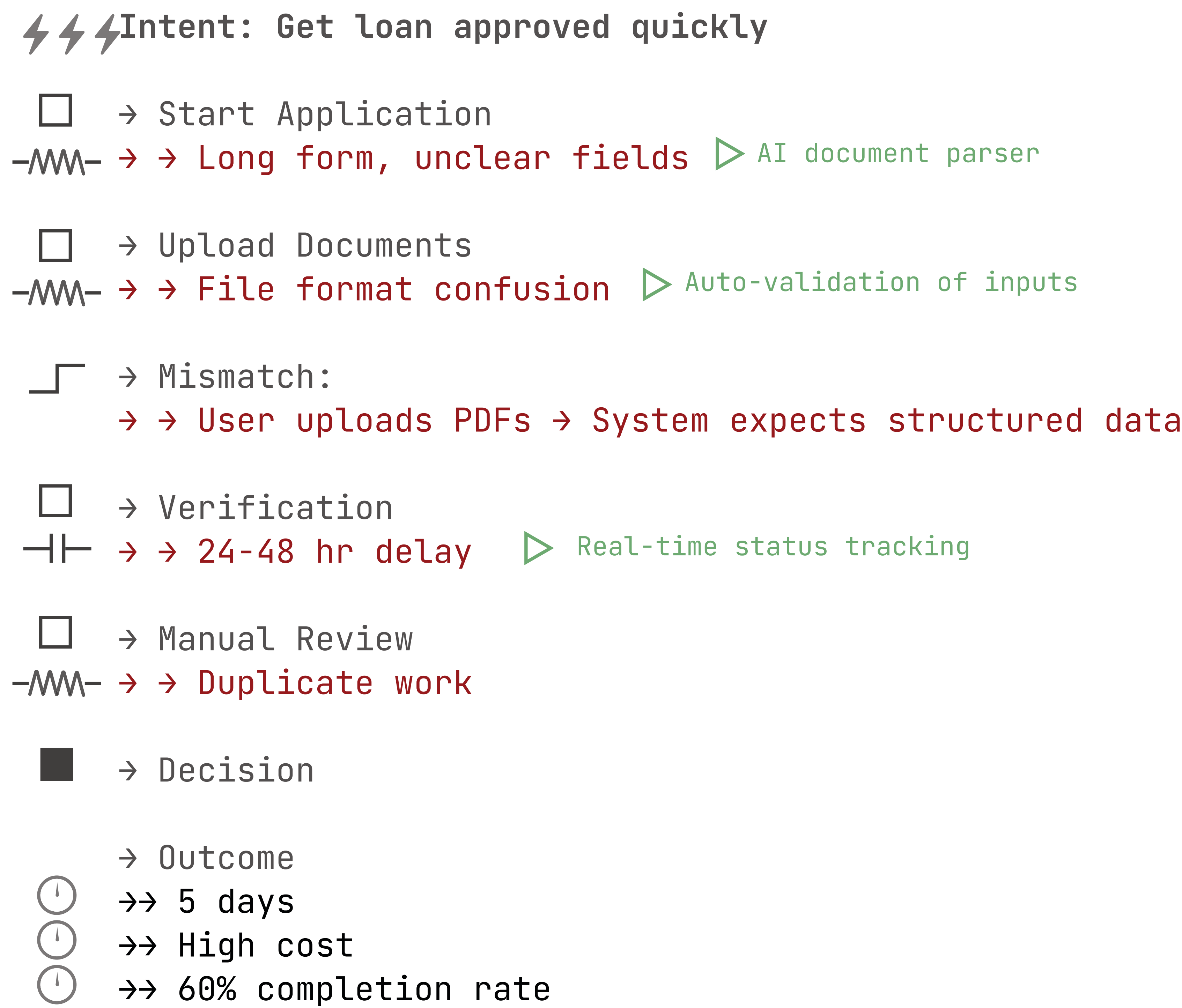
Ask:

“Where is energy being lost in the system?”

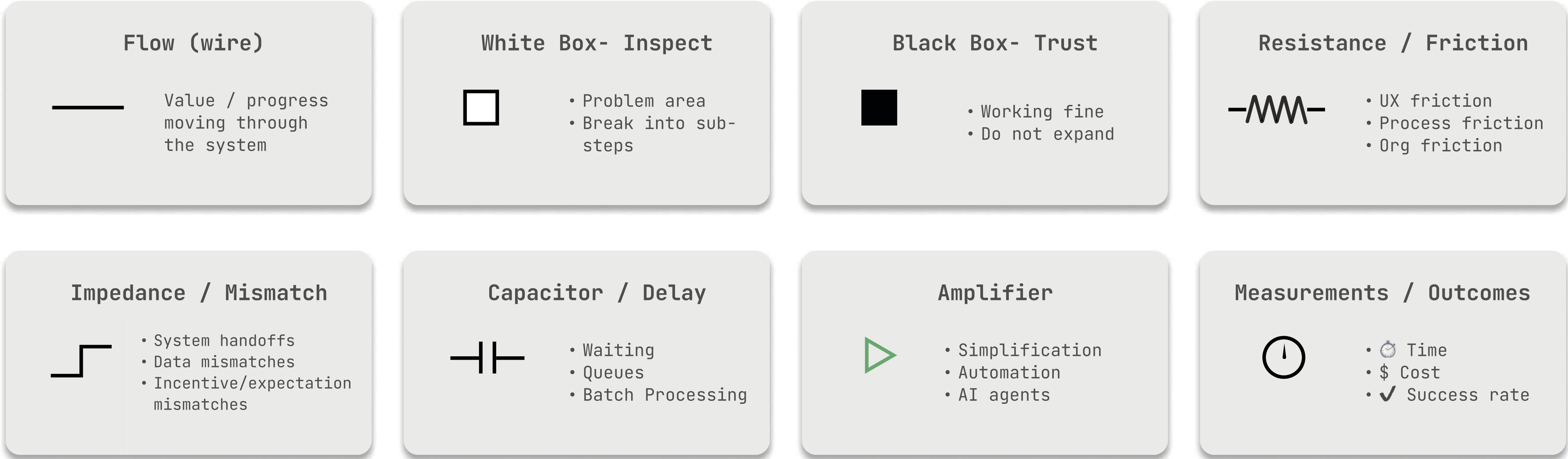
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EXAMPLE: SMALL BUSINESS LOAN APPLICATION



⚡ CIRCUIT COMPONENTS ⚡



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HOW TO BUILD (STEP-BY-STEP)

Facilitation Questions

1. DEFINE INTENT (VOLTAGE) ⚡

What is the user trying to achieve?

- What job is the user renting the process for?
- What is the intensity of the intent?

2. DRAW THE FLOW

5-9 steps from intent → outcome

- What did the end-user describe as their experience?
- What’s happening beneath the hood?

3. ADD RESISTANCE

Where is it hard, slow, or frustrating?

- Where do users struggle?
- Where do teams to extra work?

4. MARK MISMATCHES

Where do things not align?

- Where do systems not align ?
- Where do expectations break?

5. ADD DELAYS

Where does time accumulate?

- Where are there queues, buffers or delays?
- What waits on what?

6. OPEN ONE WHITE BOX

Zoom into the biggest problem area

- Which processes/steps have the most opportunity?
- Consider existing research where needed

7. ADD AMPLIFIERS

Where can we reduce loss or increase flow?

- What can be simplified?
- What can be automated?
- What can be made intelligent?

8. MEASURE OUTCOMES

What improves?

- What is being measured?
- What marks success?

$$\text{Outcome Quality} \propto (\text{Flow} - \text{Resistance} - \text{Mismatch} - \text{Delay}) + \text{Amplification}$$

THE UNLOCK 🔓

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